

Two Pages Report for the Mini-Project

- **Question & Why It Matters:-**

- “Which thing is more affected to be patient for Heart Disease? & can we prove it ?”
- Heart disease and heart attacks are major health concerns.
This analysis will identify which factors are most strongly associated with heart disease and test whether these relationships are statistically significant.

- **Data & Cleaning**

- Missing numerical values were replaced using the median.
- Missing categorical values were replaced using the mode.
- Data types and categorical values were checked and corrected where required.
- Outliers were detected using the $1.5 \times \text{IQR}$ method. Potential outliers were found in age, blood pressure, cholesterol, thalach, oldpeak, maximum heart-rate reserve, and risk score.
- Medically plausible outliers were retained rather than automatically removed.
- Duplicate records were identified but retained because there was no unique patient identifier to confirm whether they were invalid duplicates.
- After cleaning, the dataset contained 1,035 records, 16 variables, and 0 missing values.

- **EDA**

- The dataset was explored using **univariate, bivariate, and multivariate analysis**. Univariate analysis examined the distributions of key variables such as age, blood pressure, cholesterol, and heart disease status. Bivariate analysis compared categorical and numerical variables with the heart disease outcome. Multivariate analysis was performed using a correlation heatmap to examine relationships among numerical variables. Overall, **chest pain type showed a noticeable relationship with heart disease**, while most numerical variables showed relatively weak correlations with the target.

- **Hypothesis Testing**

- **Hypothesis 1 — Chest Pain Type**

H₀: Chest pain type and heart disease are independent.

H₁: Chest pain type and heart disease are not independent.

Test: Chi-square test

$\chi^2 = 260.62$, $p < 0.001$

Conclusion: Reject H₀. Chest pain type is significantly associated with heart disease.

- **Hypothesis 2 — Cholesterol**

H₀: Mean cholesterol is the same for both heart-disease groups.

H₁: Mean cholesterol is different between the groups.

Test: Independent t-test

$t = 3.54$, $p = 0.00042$

Conclusion: Reject H₀. Mean cholesterol levels are significantly different between patients with and without heart disease.

Key Findings

- **Chest pain type** showed a significant association with heart disease.
- **Cholesterol levels** were significantly different between the two heart-disease groups.
- Most other numerical variables showed **weak relationships** with heart disease.

Limitations

The analysis shows **association, not causation**. Outliers and retained duplicate records may also influence the results.